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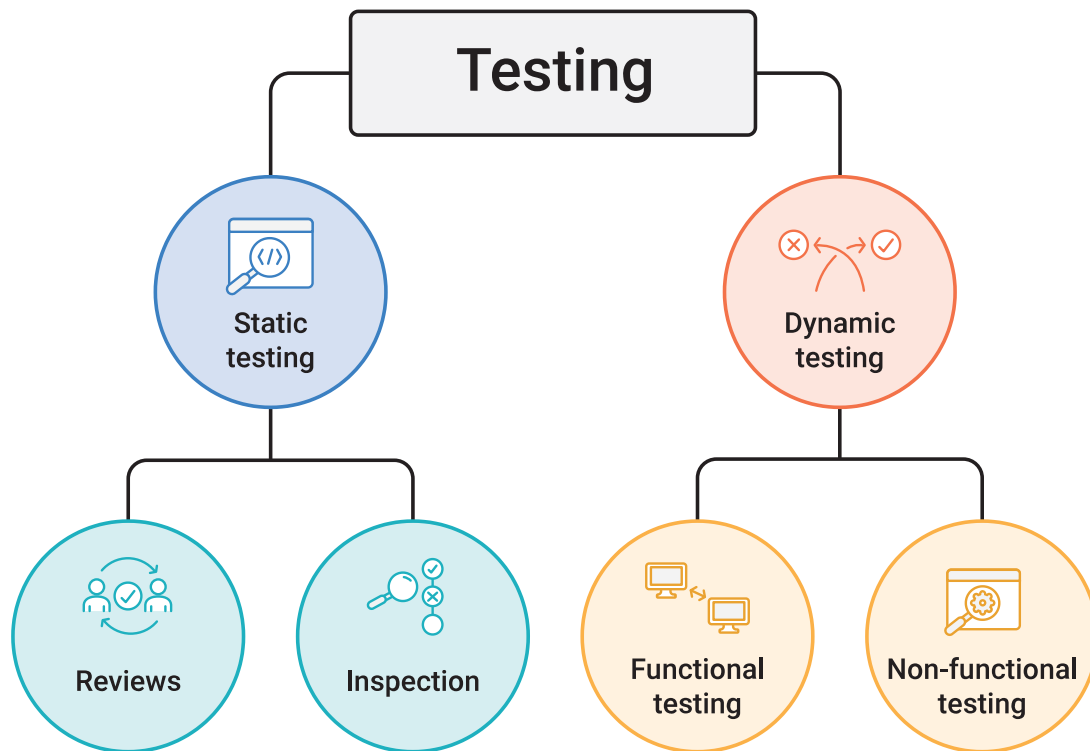
Static and dynamic analysis 15m

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Static and dynamic testing

 Click on each of the circles under Testing for more information.



Static testing techniques

Static testing techniques—also known as document-based testing techniques—do not involve executing the code or running tests on the application under test (AUT). Instead, they are designed to identify issues early in the development lifecycle, such as: missing

requirements

- design specification issues/defects
- interface problem defects
- gaps in functional or nonfunctional requirements.

Documentation reviews are often performed through formal walkthroughs with the project manager, business representatives, business analysts, developers and other stakeholders. Reviews are performed across documents such as the following:

- technical and functional specifications
 - design documents
 - code documents
 - test strategies
 - test plan documents
-
- support and maintenance documents.

These reviews are conducted at any stage, but thorough and early review of the documentation facilitates early defect detection, reducing rework costs. Formal reviews have the added bonus of improving the quality of subsequent documentation.

Static analysis is a form of white-box testing technique, as it involves understanding the internal design of a system.



☒ Static analysis

Non-Assessed

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What is the best example of static analysis

- ☐ The analysis of batch programs
- ☐ The reviewing of test plans
- ☐ The analysis of program code or other software artifacts
- ☐ The use of black-box testing

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Dynamic testing

As the name suggests, dynamic testing checks for runtime functional behaviour of a software system, including overall performance of a system and memory/CPU usage. Under dynamic testing, a code is executed or validated. This type of testing focuses on the behaviour of the code and confirms that the product works correctly based on the business requirements.

It's completed by a tester, who executes test cases against a product or program to validate the processes and confirm the product performs as expected.

This form of testing is performed in the later stages of development whilst writing code and running it to test.

Dynamic testing may involve both black-box and white-box test techniques. It's conducted throughout these test levels:

- unit testing – developers testing the source code early
- integration testing – testers testing each interface affected by the change
- system testing – testers testing the whole solution

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- acceptance testing – testers or end-users testing from a user acceptance perspective or from a business end-to-end view.

The purpose of dynamic testing is to validate the product and confirm the input and output of information within the system are correct.



Thoughts on this page?

